idea is to represent and manipulate *polynomials* by only referring to their *coefficients*. The question is, how do we calculate A3[ ], given that A1[ ], A2[ ], and A3[ ] all represent polynomials? The answer is that when two polynomials are multiplied, their coefficients are *convolved*. In equation form: A1[ ] ∗ A2[ ] = A3[ ]. This allows a standard convolution algorithm to find the transfer function of cascaded stages by convolving the two numerator arrays and the two denominator arrays.

The procedure for combining parallel stages is slightly more complicated. In algebra, fractions are added according to:

$$\frac{w}{x} + \frac{y}{z} = \frac{w \cdot z + y \cdot z}{x \cdot z}$$

Since each of the transfer functions is a fraction (one polynomial divided by another polynomial), we combine stages in parallel by multiplying the denominators, and adding the cross products in the numerators. This means that the denominator is calculated in the same way as for cascaded stages, but the numerator calculation is more elaborate. In line 340, the numerators of *cascaded* stages are convolved to find the numerator of the combined transfer function. In line 350, the numerator of the *parallel* stage combination is calculated as the sum of the two numerators convolved with the two denominators. Line 360 handles the denominator calculation for both cases.

Spectral Inversion

Chapter 14 describes an FIR filter technique called *spectral inversion*. This is a way of changing the filter kernel such that the frequency response is flipped top-for-bottom. All the passbands are changed into stopbands, and vice versa. For example, a low-pass filter is changed into high-pass, a band-pass filter into band-reject, etc. A similar procedure can be done with recursive filters, although it is far less successful.

As illustrated in Fig. 33-5, spectral inversion is accomplished by subtracting the output of the system from the original signal. This procedure can be

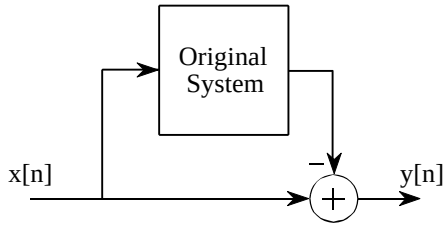

FIGURE 33-5
Spectral inversion. This procedure is the same as subtracting the output of the system from the original signal.